MILITARY SPECIFICATION

TOW 2 FUZE PROBE

This specification is approved for use by the U.S Army Armament Research and Development Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the requirements, quality assurance provisions, testing and packaging of a fuze probe for use in the TOW 2 Guided Missile.

2. APPLICABLE DOCUMENTS

2.1 Government documents

2.1.1 Specifications and standards. Unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions for

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedure and Tables for Inspection by Attributes

FSC 1336

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

DRAWINGS

U. S. Army Armament Research and Development Command

9342171 TOW 2 Fuze Probe

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.1.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Materials. Materials shall be in accordance with the applicable drawings and specifications.

3.2 Components and assemblies. The components and assemblies shall comply with all the requirements specified on Drawing 9342171 and associated drawings and with all requirements specified in applicable specifications and standards.

3.3 Item definition. The TOW 2 Fuze Probe is an extendable device for increasing warhead standoff. A crush switch is provided at the forward tip of the probe and at the ogive immediately forward of the warhead. The probe is mechanically locked in the stowed position prior to launch and extends immediately after launch. The probe extends by use of springs and is actuated by a bore rider.

3.4 Angular alignment. The angle formed between a point tangent to the probe at 1.5 probe diameters aft of the probe tip and the warhead centerline shall not exceed 40 minutes total included angle with a load simulating the flight conditions of 23.9 pounds applied to the probe tubes.

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3.5 Deployment. The time from bore rider release to probe full extension shall not exceed 0.100 seconds and the forward tip of the fully extended and locked probe shall be 16.375 inches minimum from the assembly base. The bore rider shall be free to return to the retracted position after deployment.

3.6 Electrical capacitance. The capacitance as measured between the brown and black wires of the Fuze Probe shall be 96±25 Picofarads in the stowed position and 95±35 Picofarads in the deployed position.

3.7 Electrical resistance.

3.7.1 The electrical resistance shall not exceed 1 Ohm on each of the following:

a. black wire to outer ogive base.

b. black wire to outer probe tip.

c. brown wire to inner probe tip.

d. brown wire to inner ogive.

3.7.2 The electrical resistance between the brown and black wires shall be 100 megohms minimum at 100 ± 5 Vdc.

3.8 Flight vibration. The Fuze Probe shall be subjected to the Flight Vibration Test.

3.9 Launch shock. The Fuze Probe shall be subjected to the Launch Shock Test.

3.10 Transportation vibration. The Fuze Probe shall be subjected to the Transportation Vibration Test.

3.11 Workmanship. The Fuze Probe and its components shall be fabricated and finished in a thorough, workmanlike manner and shall be free of burrs, chips, sharp edges, unblended radii, cracks, surface defects, dirt, grease, rust, corrosion products and other foreign matter. The cleaning methods used shall not be injurious to any part, nor shall parts be contaminated by the cleaning agents. All required markings shall be clearly ledgible.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspections & standard quality assurance program. Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detailed specification.

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4.2 Classification of inspection. In conformance with MIL-A-48078, the following types of inspection shall be conducted on this item:

a. First Article Inspection

b. Quality Conformance Inspection

4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a First Article sample as designated by the Contracting Officer. The First Article sample shall consist of 28 Fuze Probes and 10 each of every component and 10 each of every subassembly.

4.3.2 Inspection to be performed. See MIL-A-48078.Examine and test the 28 First Article Fuze Probes in accordance with 4.4.2.1 and TABLE I

4.3.3 Rejection See MIL-A-48078

4.4 Quality conformance inspection

4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provision of MIL-A-48078.

4.4.2 Examination MIL-A-48078, paragraph 4.4.2, applies. Unless otherwise specified in the Classification of Defects and Test Tables, sampling plans and procedures for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-T- 63439 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.1	TOW 2 Fuze Probe Assembly				9342171
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Probe deployment at ambient temperature		100%	3.5	4.5.2
102	Electrical capacitance *		100%	3.6	4.5.3
103	Electrical resistance*		100%	3.7	4.5.4
104	Weight		1.0%		Gage
105	Gap between Probe Assy & Rear of Safety Lock Knob		1.0%		Gage
Minor					
201	Thread locking compound missing		4.0%		Visual
202	Marking missing or illegible		4.0%		Visual
203	Workmanship		1.0%		Visual

NOTES:

* Electrical checks to be conducted in the stowed position after probe is deployed and restowed.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-T- 63439 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.2	Frame And Release Mechanism Assy				9342172
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Rivet missing		1.0%		Visual
Minor					
201	Workmanship		1.0%		Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-T- 63439 (AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.3	Release Mechanism Assembly		9342174
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Stake shall support 2 lbs. unseating force without separation		1.0%		Gage
102	End Shake Adequate		1.0%		Gage
103	Evidence of lubricant		1.0%		Visual
Minor					
201	Workmanship		1.0%		Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-T- 63439 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.4	Frame				9342176
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Large O.D.		1.0%		Gage
102	Small I.D.		1.0%		Gage
103	Perpendicularity of small I.D. to datum B		1.0%		Gage
104	Angle of taper at aft end of small I.D.		1.0%		Gage
105	Location of large slot on tubular section		1.0%		Gage
106	Locations of 6 aft holes		1.0%		Gage
107	Locations of 6 aft elongated holes		1.0%		Gage
108	Size of large slot on tubular section		1.0%		Gage
Minor					
201	Wall thickness		4.0%		Gage
202	Locations of two holes for Release Mechanism Assembly		4.0%		Gage
203	Protective finish is acceptable		4.0%		Visual
204	Workmanship		1.0%		Visual

NOTES:

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MIL-T- 63439 (AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.5	Ogive		9342177
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Aft I.D.		1.0%		Gage
102	Location of large forward hole		1.0%		Gage
103	Locations of six forward holes		1.0%		Gage
104	Locations of six aft holes		1.0%		Gage
105	Locations of five rivnut holes		1.0%		Gage
106	Location of depressed hole		1.0%		Gage
107	Width of cutout		1.0%		Gage
Minor					
201	Wall thickness		4.0%		Gage
202	Diameter of large forward hole		4.0%		Gage
203	Diameters of six forward holes		4.0%		Gage
204	Diameters of 6 aft holes		4.0%		Gage
205	Diameters of 5 rivnut holes		4.0%		Gage
206	Diameter of depressed hole		4.0%		Gage
207	Depth of indentation around depressed hole		4.0%		Gage
208	Protective finish is acceptable		4.0%		Visual
209	Workmanship		1.0%		Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.6	Contact, Base	1 OF 1	9342178
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	O.D. of flange		1.0%		Gage
102	I.D.		1.0%		Gage
Minor					
201	Wall thickness		4.0%		Gage
202	Locations of four holes on tubular section		4.0%		Gage
203	Overall height		4.0%		Gage
204	Locations of four holes on flange		4.0%		Gage
205	Protective finish is acceptable		4.0%		Visual
206	Workmanship		1.0%		Visual

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CLASSIFICATION OF DEFECTS & TESTS

MIL-T-63439 (AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.7	Insulator, Base Contact		9342179
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Locations of four holes		1.0%		Gage
Minor					
201	Thickness		4.0%		Gage
202	Overall height		4.0%		Gage
203	Overall length		4.0%		Gage
204	Workmanship		1.0%		Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	OF	DRAWING NUMBER
4.4.2.8	Anchor, Spring	1	1	9342180
				NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined.				
MAJOR	None defined				
MINOR					
201	Protective finish is acceptable		4.0%		Visual
202	Hole location		4.0%		Gage
203	Hole diameter		4.0%		Gage
204	Workmanship		1.0%		Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS MIL-T-63439 (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.9	Plate, Release Mechanism	1 OF 1	9342181
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Locations of three holes in side		1.0%		Gage
102	Diameters of three holes in side		1.0%		Gage
Minor					
201	Material thickness		4.0%		Gage
202	Location of hole in flange		4.0%		Gage
203	Diameter of hole in flange		4.0%		Gage
204	Contour of Plan View		4.0%		Gage
205	Cracks at bend		4.0%		Visual
206	Protective finish is acceptable		1.0%		Visual
207	Workmanship		1.0%		Visual

NOTES:

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PARAGRAPH	TITLE	SHEET	OF	DRAWING NUMBER
4.4.2.10	Pin, Tube Lock	1	1	9342182

NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None Defined				
Major					
101	Size of hole		1.0%		Gage
102	Diameter of straight shaft		1.0%		Gage
103	Overall length		1.0%		Gage
104	Length from hole to end of straight shaft.				
Minor					
201	Finish		1.0%		Gage
202	Workmanship		1.0%		Visual

NOTES:

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MIL-T- 63439 (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.11	Post, Insulator	1 OF 1	9342186
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major 101	Dimensions Improper		1.0%		Gage
Minor 201	Workmanship		1.0%		Visual

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PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.12	Arm, Bore Rider	1 OF 1	9324187
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Location of hole with boss		1.0%		Gage
102	Diameter of hole with boss		1.0%		Gage
103	Location of slotted hole in edge		1.0%		Gage
Minor					
201	Height from hole in boss to top flat		4.0%		Gage
202	Height from hole in boss to bottom flat		4.0%		Gage
203	Contour		4.0%		Gage
204	Workmanship		1.0%		Visual

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MIL-T-63439 (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.13	Shaft, Tube Lock Pin	1 OF 1	9342189
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Large diameter		1.0%		Gage
102	Medium diameters		1.0%		Gage
103	Length of medium diameters		4.0%		Gage
Minor					
201	Small diameters		4.0%		Gage
202	Length of large diameter		4.0%		Gage
203	Protective finish is acceptable		4.0%		Visual
204	Workmanship		1.0%		Visual

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MIL-T-63439 (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.14	Insulator, Tip Contact	1 OF 1	9342190
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	O.D.		1.0%		Gage
102	Overall length		1.0%		Gage
Minor					
201	Locations of three counterbored holes		4.0%		Gage
202	Diameters of counterbored holes		4.0%		Gage
203	Locations of two interior grooves		4.0%		Gage
204	Location of one circular indentation		4.0%		Gage
205	Workmanship		1.0%		Visual

NOTES:

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PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.15	Spacer, Insulating				9342191
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None Defined				
Major	None Defined				
Minor					
201	Height of shoulder		4.0		Gage
202	Small Outside Diameter		4.0		Gage
203	Workmanship		1.0		Visual

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CLASSIFICATION OF DEFECTS & TESTS

MIL-T-63439

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.16	Shaft, Bore Rider Arm	1 OF 1	9342192
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Large diameter		1.0%		Gage
102	Medium diameters		1.0%		Gage
Minor					
201	Length of large diameter		4.0%		Gage
202	Lengths of medium diameters		4.0%		Gage
203	Protective finish is acceptable		4.0%		Visual
204	Workmanship		1.0%		Visual

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MIL-T- 63439 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.17	Sleeve,Tube Lock Pin				9342193
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None Defined				
Major					
101	Small Diameter		1.0		Gage
102	Overall Length		1.0		Gage
103	Counterbore Diameter		1.0		Gage
Minor					
201	Width of Flange		4.0		Gage
202	Slot widths		4.0		Visual
203	Finish present		4.0		Visual
204	Workmanship		1.0		Visual

NOTES:

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MIL-T- 63439 (AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.18	Carrier, Probe Extension Springs		9342195
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major	None defined				
Minor					
201	O.D. of tubular section		4.0%		Gage
202	Length of tubular section		4.0%		Gage
203	O.D. of circular flange		4.0%		Gage
204	Relative locations of three arms		4.0%		Gage
205	Width of arms		4.0%		Gage
206	Wall thickness		4.0%		Gage
207	Protective finish is acceptable		4.0%		Visual
208	Workmanship		1.0%		Visual

NOTES:

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MIL-T- 63439 (AR)

PARAGRAPH	TITLE	SHEET	OF	DRAWING NUMBER
4.4.2.19	Spacer, Release Mechanism	1	1	9342196
				NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major	None defined				
Minor					
201	Diameters of two end sections		4.0%		Gage
202	Diameter of center section		4.0%		Gage
203	Length of center section		4.0%		Gage
204	Workmanship		1.0%		Visual

NOTES:

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MIL-T- 63439 (AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.20	Rivet, Drive Nylon		9342198
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major	None defined				
Minor					
201	Material Correct		4.0%		Visual
202	Workmanship		1.0%		Visual

NOTES:

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MIL-T-63439

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.21	Spring, Probe Extension	1 OF 1	9342201
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Direction of helix		1.0%		Visual
Minor					
201	Wire diameter		4.0%		Gage
202	Spring diameter		4.0%		Gage
203	Number of coils		4.0%		Gage
204	Extended length inside ends without loss of coil contact		4.0%		Gage
205	Max. initial tension		4.0%		Gage
206	Protective finish is acceptable		4.0%		Visual
207	Workmanship		1.0%		Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.22	Tube, Outer		9342202
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Forward inner groove diameter		1.0%		Gage
102	Location of elongeated hole		1.0%		Gage
103	Size of elongeated hole		1.0%		Gage
104	Angle of taper at aft end		1.0%		Gage
105	Tube O.D.		1.0%		Gage
Minor					
201	Overall length		4.0%		Gage
202	Tube I.D.		4.0%		Gage
203	Chamfer present at forward edge		4.0%		Visual
204	Protective finish present and acceptable		4.0%		Visual
205	Workmanship		1.0%		Visual

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CLASSIFICATION OF DEFECTS & TESTS

MIL-T- 63439 (AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.23	Tube, Inner		9342203
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Tube O.D.		1.0%		Gage
102	Angle of taper at aft end		1.0%		Gage
103	Radius of forward end		1.0%		Gage
104	Thickness of forward end		1.0%		Gage
105	Location of Large hole		1.0%		Gage
106	Size of Large hole		1.0%		Gage
Minor					
201	Overall length		4.0%		Gage
202	Locations of 2 holes at forward end		4.0%		Gage
203	Tube I.D.		4.0%		Gage
204	Countersinks present		4.0%		Visual
205	Protective finish is acceptable		4.0%		Visual
206	Workmanship		1.0%		Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-T-63439 (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.24	Ring Retaining, Spring Carrier	1 OF 1	9342209
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major	None defined				
Minor					
201	Ring O.D.		4.0%		Gage
202	Thickness		4.0%		Gage
203	Protective finish is acceptable		4.0%		Visual
204	Workmanship		1.0%		Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-T-63439 (AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.25	Button, Bore Rider		9342214
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major	None defined				
Minor					
201	Thickness through center		4.0%		Gage
202	Diameter of hole		4.0%		Gage
203	Disc Thickness		4.0%		Gage
204	Button O.D.		4.0%		Gage
205	Radius of outer surface		4.0%		Gage
206	Workmanship		1.0%		Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-T-63439 (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.26	Spring, Tube Assembly	1 OF 1	9342218
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Total number of coils		1.0%		Gage
102	O.D. Compressed		1.0%		Gage
103	I.D. Relaxed		1.0%		Gage
Minor					
201	Wire diameter		4.0%		Gage
202	Free length		4.0%		Gage
203	Protective finish is acceptable		4.0%		Visual
204	Direction of Helix		4.0%		Visual
205	Workmanship		1.0%		Visual

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CLASSIFICATION OF DEFECTS & TESTS

MIL-T-63439 (AR)

PARAGRAPH	TITLE	SHEET	OF	DRAWING NUMBER
4.4.2.27	Tip Contact Assembly	1	1	9342222
				NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Parts improperly mated		1.0%		Visual
Minor					
201	Workmanship		1.0%		Visual

NOTES:

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PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.28	Wire Assembly				9342223
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None				
Major	None				
Minor					
201	Length		4.0		Gage
202	Joint strength		4.0		Gage
203	Workmanship		1.0		Visual

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MIL-T- 63439 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.29	Tube Assembly				9342224
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Parts mated improperly		1.0%		Visual
Minor					
201	Workmanship		1.0%		Visual

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.30	Wedge, Intermediate Tube	1 OF 1	9342233
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Taper angle		1.0%		Gage
102	Largest thickness		1.0%		Gage
103	Overall length		1.0%		Gage
Minor					
201	Presence of slot		4.0%		Visual
202	Protective finish is acceptable		4.0%		Visual
203	Workmanship		1.0%		Visual

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CLASSIFICATION OF DEFECTS & TESTS

MIL-T-63439 (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.31	Spring, Tube Lock Pin	1 OF 1	9342234
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major	None defined				
Minor					
201	Wire Diameter		4.0		Gage
202	O.D.		4.0		Gage
203	Min Load		4.0		Gage
204	Free length		4.0		Gage
205	Presence of Finish		4.0		Visual
206	Workmanship		4.0		Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-T-63439 (AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.32	Ring, Stop (Outer Tube)		9342237
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major	None defined				
Minor					
201	Ring I.D.		4.0%		Gage
202	Thickness		4.0%		Gage
203	Protective finish is acceptable		4.0%		Visual
204	Workmanship		1.0%		Visual

NOTES:

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MIL-T- 63439 (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.33	Wire, Stiffener	1 OF 1	9342238
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major	None defined				
Minor					
201	Wire Dia.		4.0%		Gage
202	Overall length		4.0%		Gage
203	I.D. of loop		4.0%		Visual
204	Protective finish is acceptable		4.0%		Visual
205	Workmanship		1.0%		Visual

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MIL-T- 63439 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.34	Contact, Tip				9342239
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Cone O.D.		1.0%		Gage
102	Radius of forward end		1.0%		Gage
103	Overall length		1.0%		Gage
Minor					
201	Wall thickness		4.0%		Gage
202	Locations of two large holes		4.0%		Gage
203	Locations of two small holes		4.0%		Gage
204	Diameters of two large holes		4.0%		Gage
205	Diameters of two small holes		4.0%		Gage
206	Protective finish is acceptable		4.0%		Visual
207	Workmanship		1.0%		Visual

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MIL-T-63439 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.35	Wedge, Outer Tube				9342241
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Taper angle		1.0%		Gage
102	Largest thickness		1.0%		Gage
103	Overall length		1.0%		Gage
Minor					
201	Presence of slot		4.0%		Visual
202	Protective finish is acceptable		4.0%		Visual
203	Workmanship		1.0%		Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-T-63439 (AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.36	Tube, Intermediate		9342246
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Forward inner groove diameter		1.0%		Gage
102	Location of elongated hole		1.0%		Gage
103	Size of elongated hole		1.0%		Gage
104	Tube I.D.		1.0%		Gage
105	Tube O.D.		1.0%		Gage
106	Angle of taper at aft end		1.0%		Gage
Minor					
201	Overall length		4.0%		Gage
202	Chamfer present at forward end		4.0%		Visual
203	Protective finish is acceptable		1.0%		Visual
204	Workmanship		1.0%		Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-T- 63439 (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.37	Retainer, Tube Assembly Spring	1 OF 1	9342263
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major	None defined				
Minor					
201	O.D. of flange		4.0%		Gage
202	O.D. of tubular section		4.0%		Gage
203	Overall length		4.0%		Gage
204	Wall thickness		4.0%		Gage
205	Protective finish is acceptable		4.0%		Visual
206	Workmanship		4.0%		Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.38	Ogive Assembly	1 OF 1	9342265
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None Defined				
Major	None Defined				
Minor					
201	Torque of rivnut		1.0		Gage
202	Workmanship		1.0		Visual

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.39	Safety Lock Assembly				9342266
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None Defined				
Major	None Defined				
Minor					
201	Presence of parts		4.0		Visual
202	Pushout force		1.0		Gage

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PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.40	Inner Tube Assembly		9342267
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None Defined				
Major					
101	Flathead screw flush or below		1.0		Gage
102	Blind Rivets flush or below		1.0		Gage
103	Torque on Flathead Screw		1.0		Gage
Minor					
201	Workmanship		1.0		Visual

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.41	Base Contact Assy				9342268
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None Defined				
Major	None Defined				
Minor					
201	Presence of Parts		1.0		Visual
202	Rivet installation proper		4.0		Visual

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.42	Knob, Safety Lock	1 OF 1	9342269
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Diameter of hole		1.0%		Gage
Minor					
201	O.D. of knurled area		4.0%		Gage
202	Workmanship		1.0%		Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-T- 63439 (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.43	Shaft, Safety Lock	1 OF 1	9342270
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Diameter of knurled section		1.0%		Gage
Minor					
201	Overall length		4.0%		Gage
202	Diameter of section adjacent to threads		4.0%		Gage
203	Length from stop to tapered end		4.0%		Gage
204	Thread size		4.0%		Gage
205	Protective finish is acceptable		4.0%		Gage
206	Workmanship		1.0%		Visual

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4.4.3 Testing. Lot acceptance testing shall be performed in the sequence shown in Table I on a quantity of twenty eight fuze probes. This quantity is valid for lot sizes up to 5000 units. All inspection shall be classified Major unless otherwise specified. Any item failing to meet the requirements during 100 percent testing shall be classified defective and removed from the lot. Any item failing to meet the requirements during Lot Acceptance Testing shall cause the lot to be rejected.

TABLE I - LOT ACCEPTANCE TESTING

Sample Nos.	Transportation Vibration	Launch Shock	Deployment	Flight Vibration	Angular Alignment	Electrical Capacitance & Resistance
1-2			X		X	X
3-4			C		X	X
5-12			C			X
13-14			H		X	X
15-22			H			X
23-24	X	X	X	X	X	X
25-28	X	X	X	X		X

ABBREVIATIONS: X = Test at Ambient Temperature
C = Test at Cold Temperature (-25°F)
H = Test at Hot Temperature (+140°F)

4.4.4 Inspection Equipment. The inspection equipment required to perform the examinations and tests prescribed herein is described in the "Paragraph Reference/Inspection Method" column in the tables starting with Paragraph 4.4.2.1. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078 and 6.3 herein.

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4.5 Test methods.

4.5.1 Angular alignment. Apply loads to a deployed unit to the sections as defined in Figure 3. Measure concentricity as defined in Figure 3. The load shall be applied perpendicularly to the warhead centerline in the following order: At 0°, 180°, 90° and 270°. Measure deflection and calculate probe angle of deflection.

4.5.2 Deployment. A stowed unit shall be temperature conditioned as required for a minimum of 4 hours and tested within 2 minutes of removal from the conditioning chamber. (Low temperature units may be bagged to prevent frosting). The bore rider shall be released to a point 1/4 inch from the surface of the ogive. The unit shall not deploy. The bore rider shall then be released and the probe shall deploy as required. Apply 20 lbs. longitudinally to the smallest tube section in the aft direction. (Figure 1 defines the appropriate axes). The probe shall remain locked.

4.5.3 Electrical capacitance. Measure the electrical capacitance in the deployed position.

4.5.4 Electrical resistance. Measure the electrical resistance, in the deployed position.

4.5.5 Flight vibration. Attach a deployed unit to the test equipment and apply vibration sequentially along each of three orthogonal axes. The acceleration power spectral density for each axis as measured on the probe/fixture interface shall be in accordance with Figures 2A, 2B and 2C (Figure 1 defines the appropriate axes). During vibration the brown/black leads shall be monitored for switch closures of .001 sec at 100 Vdc potential. The unit shall not be visibly damaged. After vibration apply a 20 lb. load longitudinally to the smallest tube section in the aft direction. The unit shall remain in the deployed and locked position.

4.5.6 Launch shock. Attach a stowed unit to the test equipment and apply a 400 g ± 10% 1/2 sine wave pulse for 10 ± 3 milliseconds. The axis of impact shall pass through the longitudinal axis of the unit under test with loads imposed in the direction of launch (Figure I defines the appropriate axes). The unit shall not be visibly damaged.

4.5.7 Transportation vibration. A stowed unit shall be attached to the test equipment. Vibration input, as defined by Table II, shall be imposed in each of the specified axes for 2.0 hours (Figure 1 defines the appropriate axes). The unit shall not be visibly damaged

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TABLE II

TRANSPORTATION VIBRATION

Longitudinal Axis		Vertical Axis		Transverse Axis	
Frequency Hz	Random Vibration g2/Hz	Frequency Hz	Random Vibration g2/Hz	Frequency Hz	Random Vibration g2/Hz
17	0.02	18	0.145	20	0.078
17-20	18 db/oct	18-20	18 db/oct	20-75	3 db/oct
20-30	0.03	20-45	0.21	75-120	0.29
30-100	-6 db/oct	45-55	-12 db/oct	120-400	-3 db/oct
100	0.003	55-175	0.10	400	0.085
		175-300	-3 db/oct		
		300	0.055		

5. PACKAGING

5.1 Preservation and packaging. Preservation and packaging shall be such as to prevent deterioration or damage during handling and shipment to the first destination.

5.2 Packing. Packing shall be in a manner to insure carrier acceptance and safe delivery at destination. Containers shall be in accordance with Uniform Freight Classification rules or regulations of other carriers applicable to the mode of transportation.

6. NOTES

6.1 Intended use. It is intended that the probe defined by this specification shall be used on the TOW 2 warhead for use in the TOW Guided Missile.

6.2 Ordering data. See MIL-A-48078, para. 6.1.

6.3 Submission of designs for approval. See 6.2.3 of MIL-A-48078. Submit equipment designs, as required to Commander, ARRADCOM, ATTN: DRDAR-QAR-I, Dover, New Jersey 07801. The letter of submittal shall state contractor, contract number, specification number, item nomenclature and classification of defects or test paragraph.

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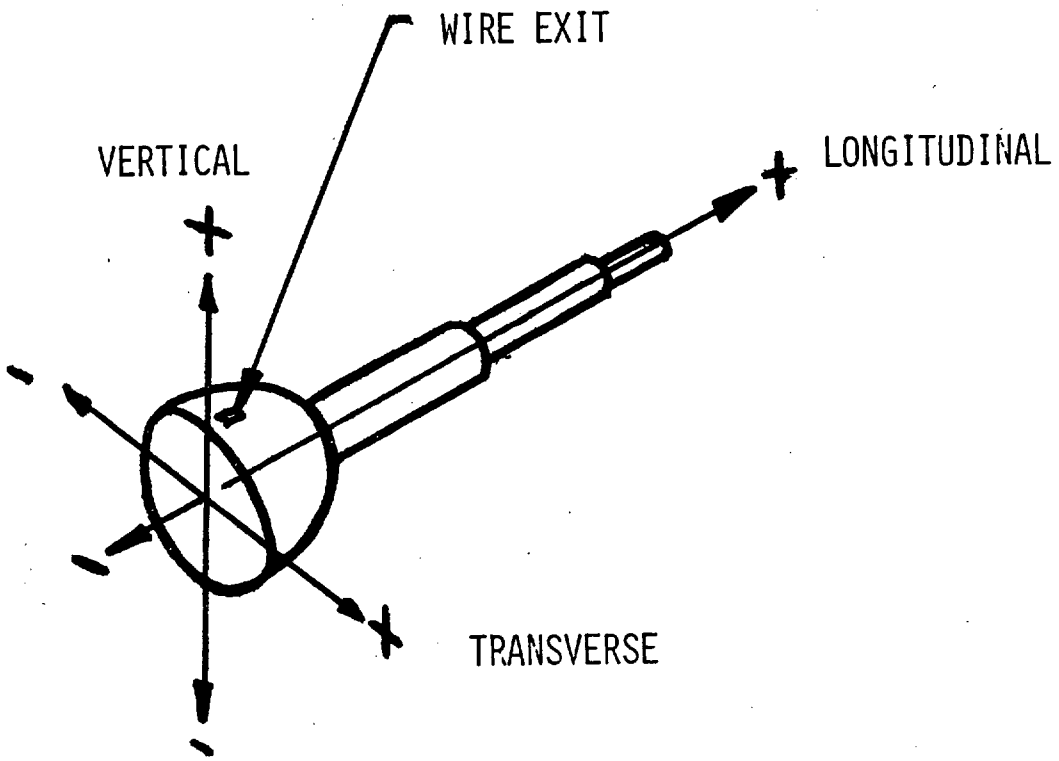
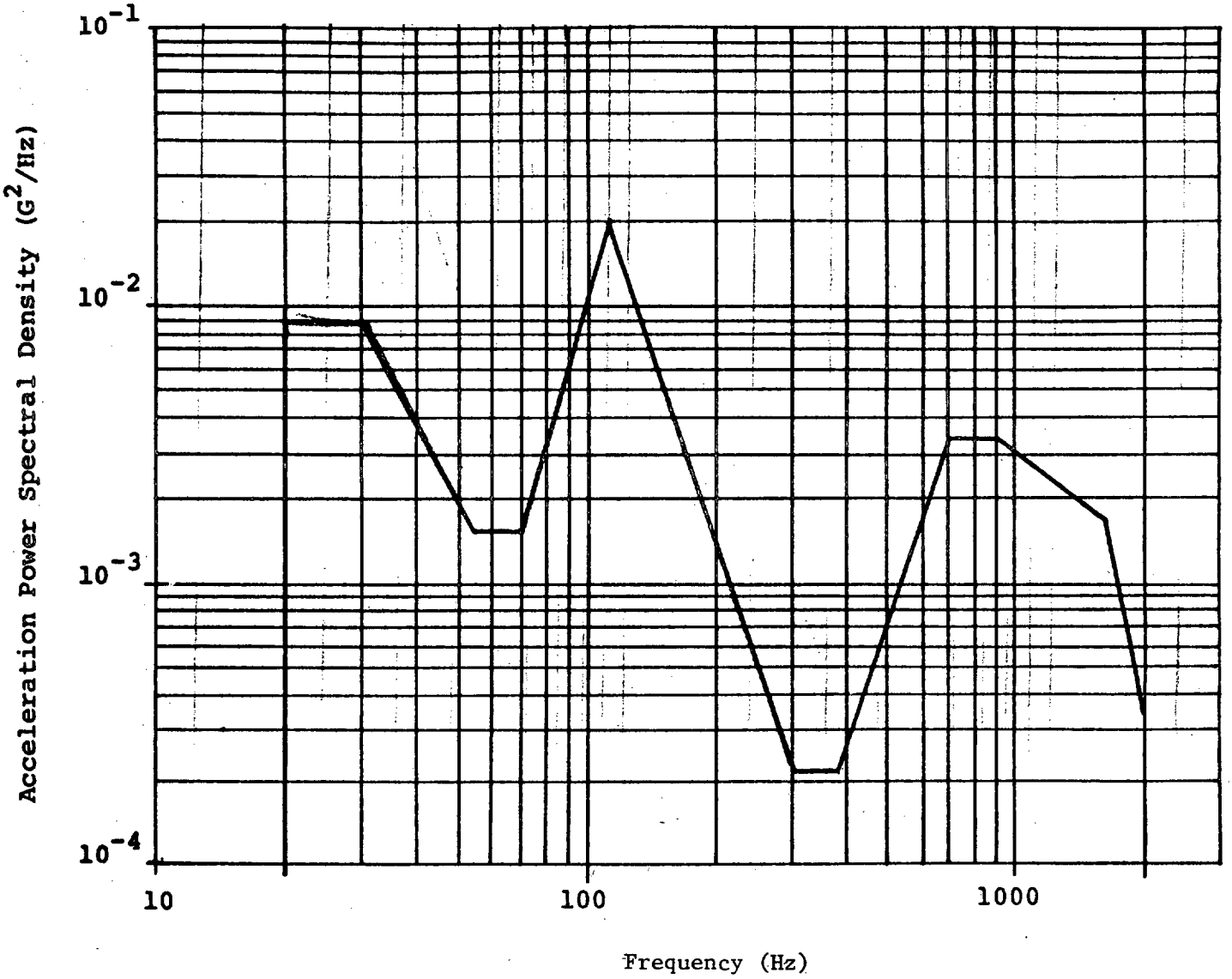


FIGURE 1. Orientation of Axes

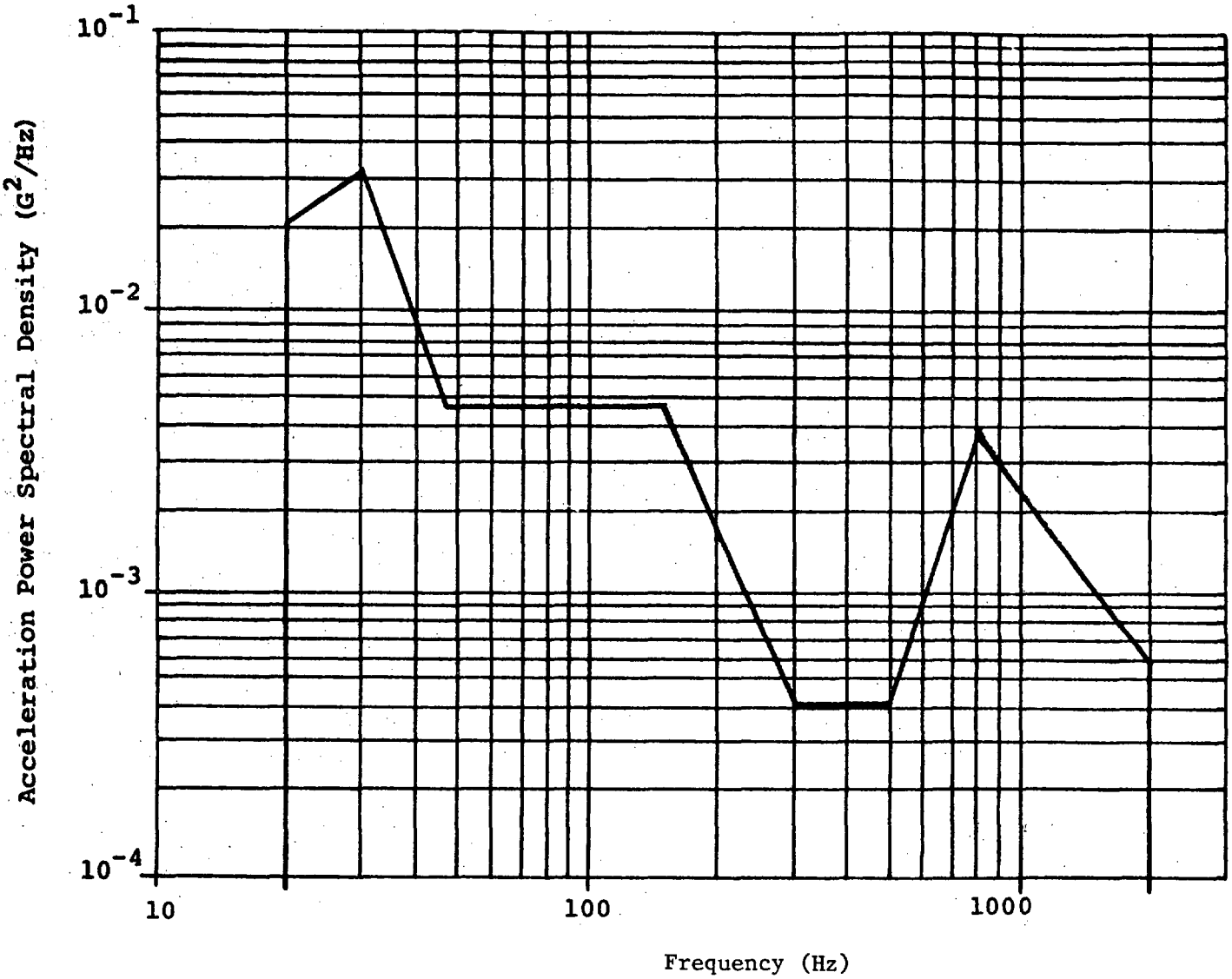
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
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Level: 2.080 G RMS.
Test Duration: 4 Sec at 2X RMS Level, followed by 20 Sec at Level.

FIGURE 2A - Transverse Axis Flight Vibration

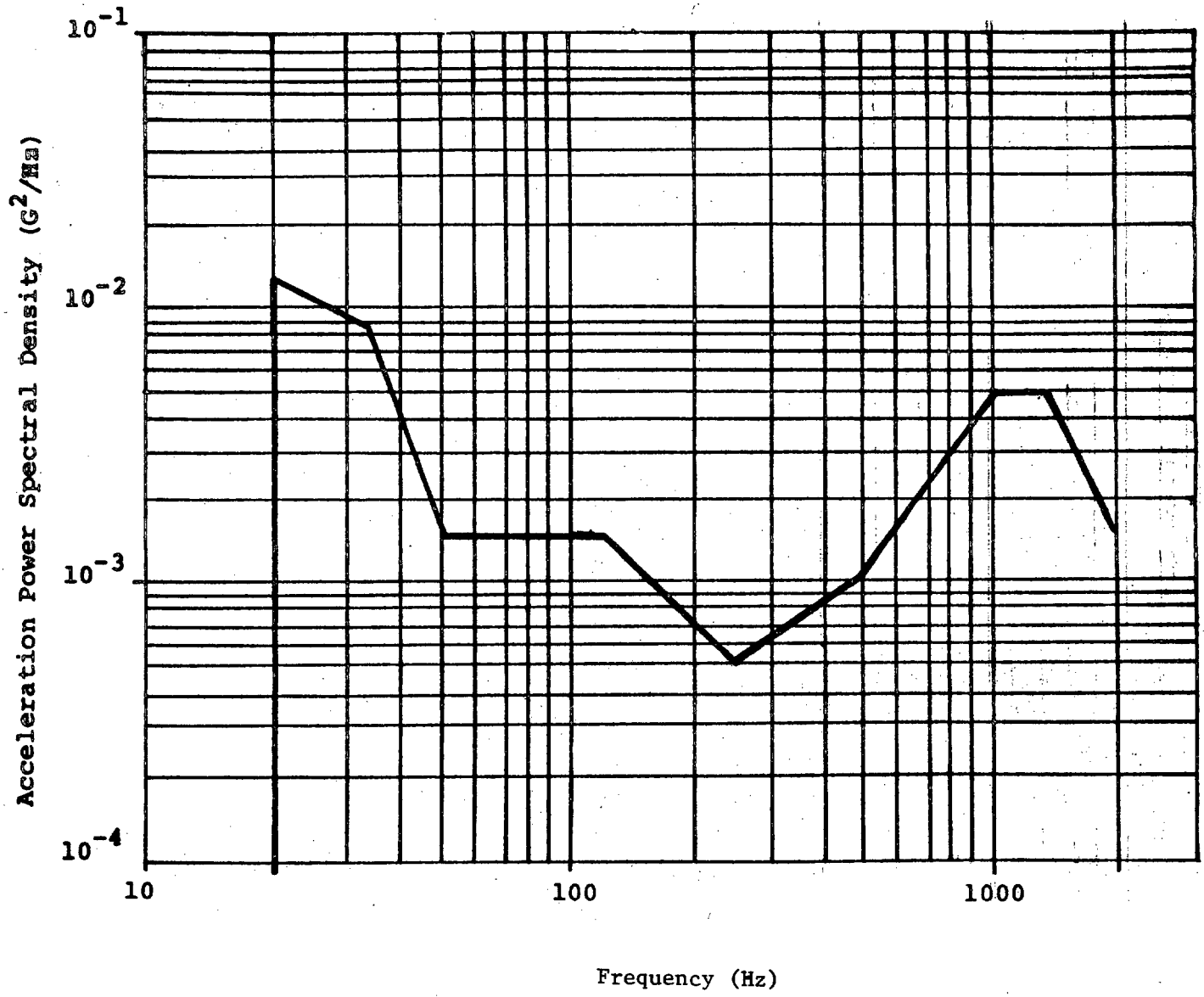
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Level: 1.900 G RMS
Test Duration: 4 seconds at 2X RMS Level,
followed by 20 Sec at Level

FIGURE 2B - Vertical Axis, Flight Vibration

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Level: 2.340 G RMS.
Test Duration: 4 seconds at 3X RMS Level, followed by 20 seconds at Level.

FIGURE 2C - Longitudinal Axis, Flight Vibration

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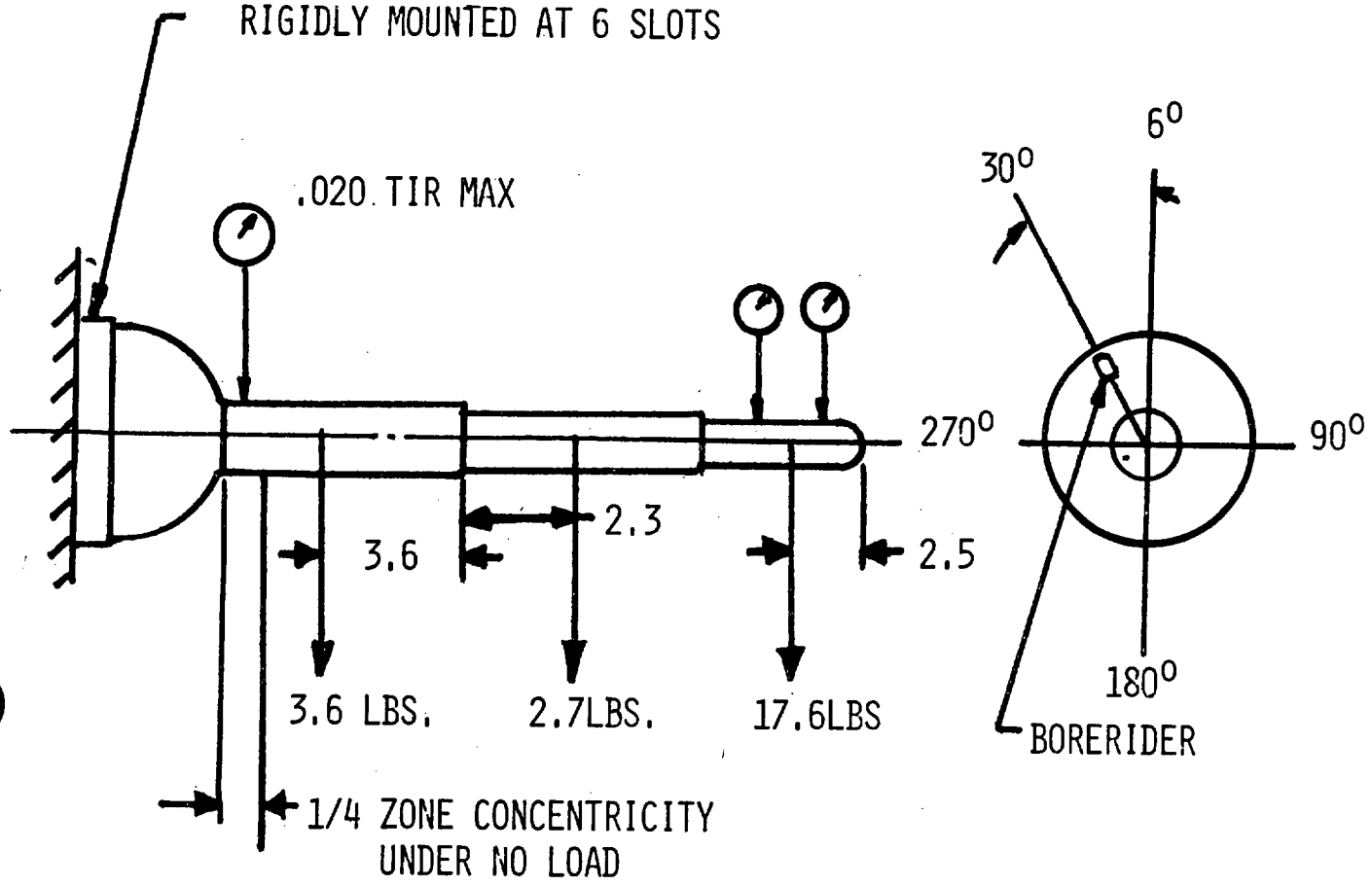


FIGURE 3 - Angular Alignment Test Load Set-up

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Preparing Activity:
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Project 1336-A385

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